EXPERIMENT - 9

AIM :

To write a Python program to solve the Travelling Salesman Problem by finding a minimum-cost tour that visits every city exactly once and returns to the starting city.

ALGORITHM:

1. Start.
2. Read the number of cities and the distance between each pair of cities.
3. Select a starting city.
4. Generate possible tours that visit every city exactly once.
5. Calculate the total distance of each tour, including the return to the starting city.
6. Compare the total distances and select the minimum-cost tour.
7. Display the optimal tour and its total distance.
8. Stop.

PROGRAM :

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```
from itertools import permutations

graph = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

n = len(graph)
cities = range(1, n)
min_cost = float("inf")
best_path = []

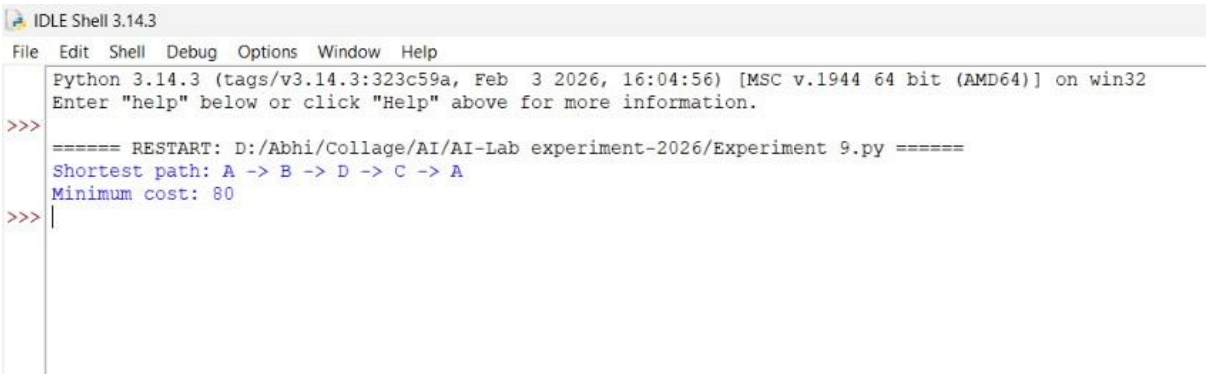
for route in permutations(cities):
    path = (0,) + route + (0,)
    cost = sum(
        graph[path[i]][path[i + 1]]
        for i in range(n)
    )

    if cost < min_cost:
        min_cost = cost
        best_path = path

names = ["A", "B", "C", "D"]

print("Shortest path:",
      " -> ".join(names[i] for i in best_path))
print("Minimum cost:", min_cost)
```

OUTPUT:



RESULT:

Thus, the Python program to solve the Travelling Salesman Problem was executed successfully, and an optimal or minimum-cost tour was obtained using the chosen algorithm.